

- 7.1 Constant field scaling allows us represent changes in the scale of CMOS devices with a unitless quantity  $S$ . Assuming ideal constant field scaling (that is, scaling the size of a CMOS device without changing the strength of its internal electric field), we may use Table 7.4 to determine that gate delay  $\tau$  scales by a factor of  $\frac{1}{S}$ . Thus, using an initial gate delay of  $\tau = 500$  ps, we may assume that by scaling the device in question by a factor of  $S = 2$ , the new gate delay  $\tau = \frac{500}{2} = 250$  ps.

Using Table 7.5, we see that repeated wire RC delay per unit length, assuming constant field scaling of gates, scales by a factor of  $\sqrt{S}$ ; in our case, this causes the delay to *increase*, so that  $t_{wr} = 500\sqrt{2} \approx 707$  ps. This concords with our intuitive sense of wire scaling and its impact on delay; reduced height in an interconnect will lower its cross-sectional area and increase its resistance, leading to higher delay.

In total, this leads to an overall reduction in delay from  $500 + 500 = 1000$  ps to  $250 + 707 = 957$  ps, or a 4.3% reduction in delay.

- 7.2 First, the worst-case input delay for the NOR2X1 gate, based on the TSMC databook,  $B \rightarrow Y \uparrow$  is 45 ps, and best-case input delay  $A \rightarrow Y \downarrow$  is 18 ps. We will use the formula given in the TSMC databook for adjusting delay based on derating factors

$$t_{TPD} = (K_{\text{process}} \cdot [1 : e + 35 + (K_{\text{volt}}) \cdot \Delta V_{dd}]) \cdot [1 + (K_{\text{temp}} \cdot \Delta T)] \cdot t_{\text{typical}}$$

and using the allowable voltage and temperature ranges found in the databook, 1.62 – 1.98 V and 0 – 125° C, respectively, we may determine worst-case and best-case delay as follows:

- (a) At  $V_{DD} = 1.62$  V and  $T = 125^\circ$  C, with a slow process speed,

$$t_{TPD} = 1.293 \cdot [1 + (-0.731 \cdot -0.18)] \cdot [1 + (0.00123 \cdot 100)] \cdot t_{\text{typical}} = 73.9 \text{ ps.}$$

- (b) At  $V_{DD} = 1.98$  and  $T = 0^\circ$  C, with a fast process speed,

$$t_{TPD} = 0.781 \cdot [1 + (-0.511 \cdot 0.18)] \cdot [1 + (0.00123 \cdot 0)] \cdot 0.018 = 12.8 \text{ ps.}$$

- 7.3 Using Eq. 7.3,

$$C = \frac{I_{\text{dc-max}}}{\alpha V_{DD} f},$$

we may solve for  $I_{\text{dc-max}}$  as

$$I_{\text{dc-max}} = C \alpha V_{DD} f.$$

Therefore,

$$I_{\text{dc-max}} = (1 \times 10^{-12})(0.1)(1.8)(1 \times 10^9) = 0.18 \text{ mA.}$$

- 7.4 (a) Process size refers to the width of the transistor's gate, and so scales by a factor of  $S$ , while gate delay scales with  $\frac{1}{S}$ . Taking the ratio of gate delay between the two processes in this way,

$$\tau_{\text{ratio}} = \frac{0.058}{0.083} \approx 0.7,$$

and so, in the 12 nm process, we may expect gate delay to be roughly 70% of gate delay in the 17 nm process.

- (b) Unrepeated RC wire delay per unit length, on the other hand, scales with  $S^2$ , and so the ratio between the two processes is,

$$t_{wu \text{ ratio}} = \frac{17^2}{12^2} \approx 2,$$

and so we may expect unrepeated RC wire delay per unit length to be roughly twice as large in the 12 nm process as in the 17 nm process.

9.3 The figure below shows a 12-input NOR gate, built using NAND and NOR gates with three inputs or less:

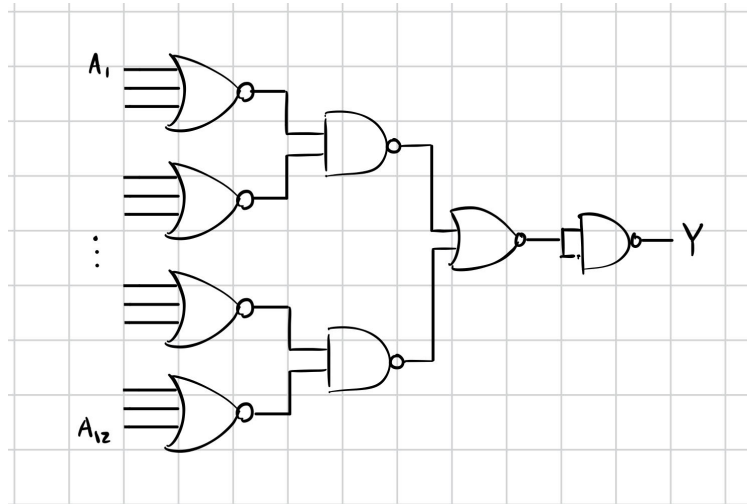


Figure 1: 12-input NOR gate built using CMOS logic

9.10 The following figure shows the layout for HI-skew and LO-skew NAND3 and NOR3 gates, sizing down transistors on the non-critical transistion by a factor of 2.

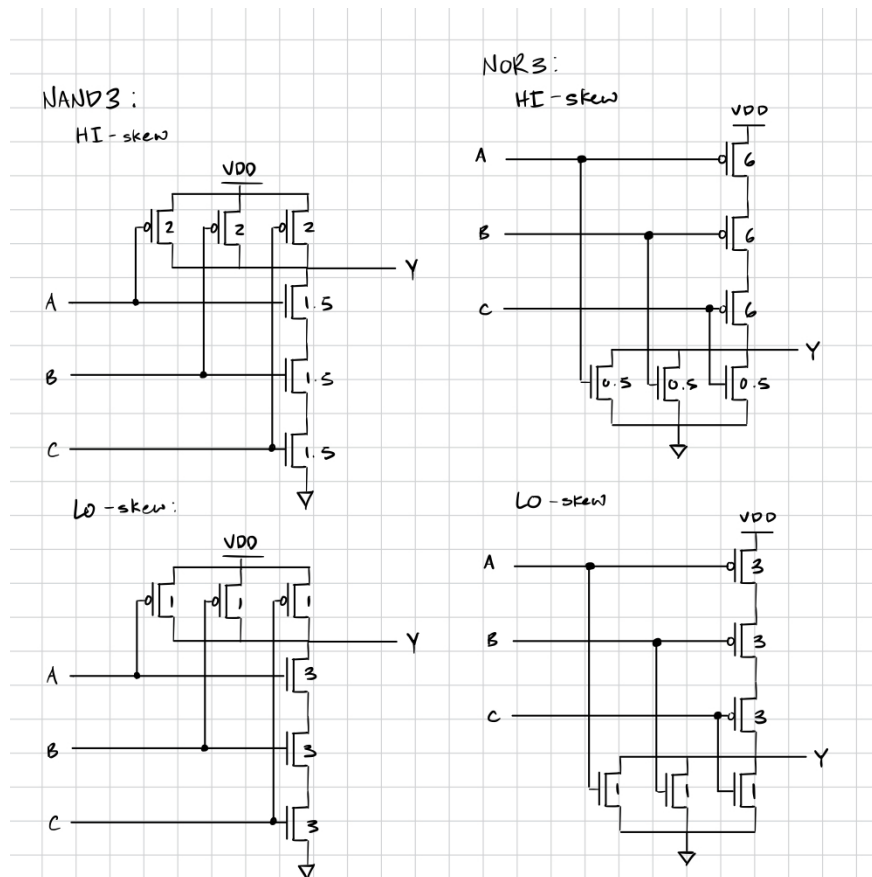


Figure 2: HI-skew and LO-skew NAND3 and NOR3 gates

Finding logical effort of each of the gates for the critical transition, the NAND3 HI-skew gate carries a logical effort of  $g_u = \frac{7}{6}$  when compared to an unskewed unit inverter capable of driving the same current; the NAND3 LO-skew gate carries a logical effort of  $g_d = \frac{4}{3}$ .

The NOR3 HI-skew gate carries a logical effort of  $g_u = \frac{13}{6}$ , and the NO3 LO-skew gate carries a logical effort of  $g_d = \frac{4}{3}$ .

- 9.18 The following figure shows the layout of a pseudo-nMOS NAND3 gate and a pseudo-nMOS NOR3 gate, with appropriately-sized transistors.

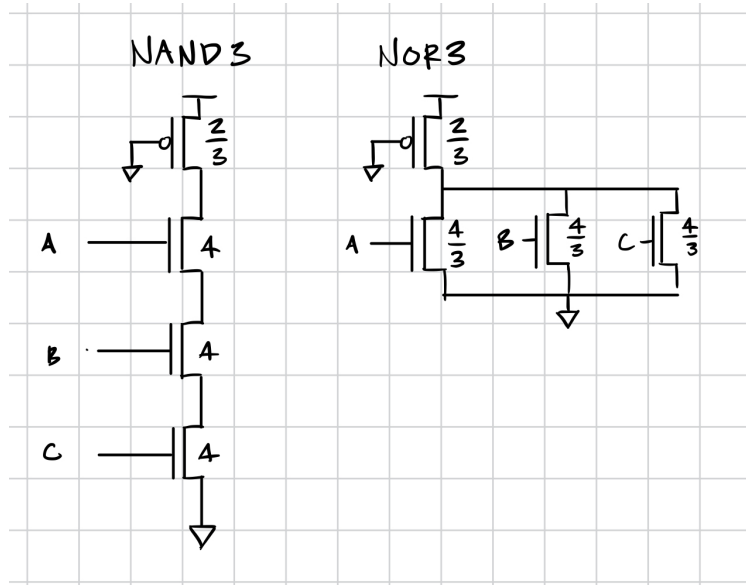


Figure 3: Pseudo-nMOS gates

For the NAND3 gate,  $g_u = 4$ ,  $g_d = \frac{4}{3}$ , and  $g_{avg} = \frac{8}{3}$ . For the NOR3 gate,  $g_u = \frac{4}{3}$ ,  $g_d = \frac{4}{9}$ , and  $g_{avg} = \frac{8}{9}$ .

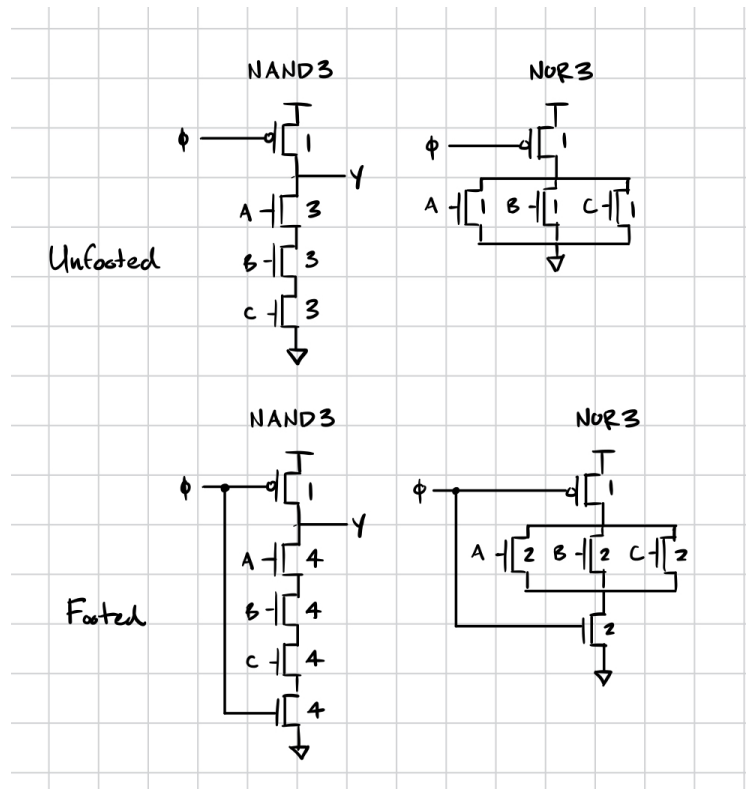


Figure 4: Dynamic footed and unfooted gates

9.27 The dynamic, unfooted NAND3 gate carries a logical effort of  $g_d = \frac{3}{3}$ , its footed counterpart a logical effort of  $g_d = \frac{4}{3}$ ; while the dynamic, unfooted NOR3 gate carries a logical effort of  $g_d = \frac{1}{3}$ , and its footed counterpart carries a logical effort of  $\frac{2}{3}$ .